



Image Outpainting

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Abstract

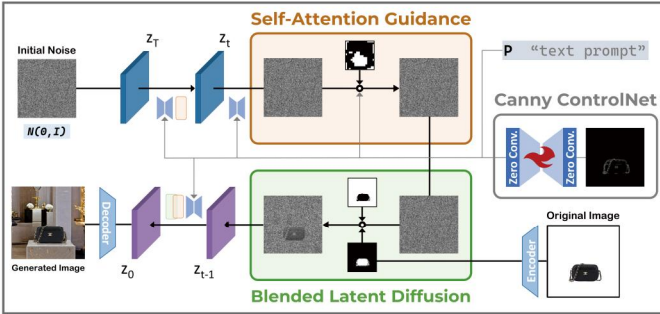
- The advertising photography market is increasingly focusing on visual storytelling.
- Our model (BLDCSAG) is designed to generate advertising photos.
- DALL-E 3 and existing models face the following challenges



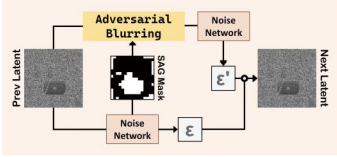
- We propose a new approach combining Blended Latent Diffusion and a Self-Attention Guidance mechanism to maintain object fidelity while generating backgrounds.

Methodology

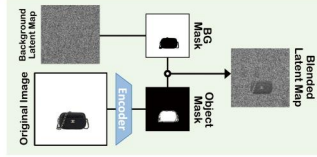
I Model Overall Architecture



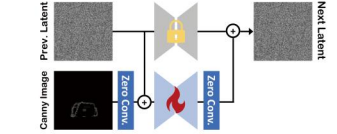
II Self-Attention Guidance



III Blended Latent Diffusion



IV ControlNet & Noise Network



V Model Pseudo Algorithm

ALGORITHM 1:
Sampling from ControlNetDiffusion Model with self attention guidance

Functions:
Model(x_t, c, d): a controlnet diffusion model that outputs the predicted noise ϵ_t , variance Σ , and self attention map A_t , given the input x_t , canny image c , text description d

Self Attention Mask(A): self-attention-based masking function
Gaussian Blur($\hat{x}$): a Gaussian blurring function.
 $\bar{u}_t$: $\bar{u}_t$ with parameter α_t given by DDIM scheduler with given timestep t

input x_t, c, d, s, t

$\epsilon_t, \Sigma, A_t \leftarrow \text{Model}(x_t, c, d)$
 $M_t \leftarrow 1(A_t > \psi)$
 $\hat{x}_0 \leftarrow (x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_t) / \sqrt{\bar{\alpha}_t}$
 $\hat{x}_0 \leftarrow \text{GaussianBlur}(\hat{x}_0)$
 $\hat{x}_t \leftarrow \sqrt{\bar{\alpha}_t}\hat{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_t$
 $\hat{x}_t \leftarrow (1 - M_t)\hat{x}_t + M_t\hat{x}_0$
 $\bar{\epsilon} \leftarrow \text{Model}(\hat{x}_t, c, d)$
 $\bar{\epsilon} \leftarrow \bar{\epsilon} + (1 + s) \times (\epsilon_t - \bar{\epsilon})$
 $x_t \sim N\left(\frac{1}{\sqrt{\bar{\alpha}_t}}\left(\hat{x}_t - \frac{1 - \bar{\alpha}_t}{\sqrt{1 - \bar{\alpha}_t}}\bar{\epsilon}\right), \Sigma\right)$

return x_t

ALGORITHM 2:
BLDCSAG given a text conditioned ControlNet Latent Diffusion model
 $(VAE = \{E(x), D(x)\}, \text{SelfAttentionGuidedControlNetDiffusionModel} = \{\text{noise}(x, t), \text{denoise}(x, c, d, s, t)\})$ with **ALGORITHM 1**

Input: source image x , background text description d , input mask m , diffusion steps k , blending start step t_{blend} , canny image c , self attention guidance scale s

Output: edited image $\hat{x}$ that differs from input image x outside area m according to text description d canny image c

$m_{latent} = \text{downsample}(m)$
 $x_{init} \sim E(x)$
 $z_k \sim \text{noise}(x_{init}, k)$
for all t **from** k **to** 0 **do**
 $z_{object} \sim \text{noise}(z_{init}, t)$
 $z_{bg} \sim \text{denoise}(z_t, c, d, s, t)$
if $t > t_{blend}$ **do**
 $z_t \leftarrow z_{bg}$
else do
 $z_t \leftarrow z_{object} \odot m_{latent} + z_{bg} \odot (1 - m_{latent})$
end for
 $\hat{x} = D(z_0)$
return $\hat{x}$

Conclusion

- We propose BLDCSAG, a powerful end-to-end image inpainting pipeline.
- To preserve the identity of objects, we integrate ControlNet and BLD.
- To reconstruct visually appealing backgrounds, we utilize SAG.
- Removing relevant object in the background should be solved.

More results

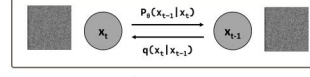


GitHub QR



Backgrounds

Denoising Diffusion Probabilistic Models



$$L_{\text{simple}}(\theta) := \mathbb{E}_{t, x_0, \epsilon} \left[\left\| \epsilon - \epsilon_{\theta}(x_0, t) \right\|_2^2 \right]$$

Latent Diffusion Models

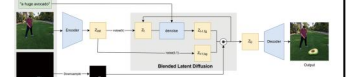
$$L_{DM} = \mathbb{E}_{x, \epsilon \sim N(0, 1), t} \left[\left\| \epsilon - \epsilon_{\theta}(x_t, t) \right\|_2^2 \right]$$

$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), y, \epsilon \sim N(0, 1), t} \left[\left\| \epsilon - \epsilon_{\theta}(z_t, t, \tau_{\theta}(y)) \right\|_2^2 \right]$$

ControlNet

$$y_c = \mathcal{F}(x; \Theta) + \mathcal{Z}(\mathcal{F}(x + \mathcal{Z}(c; \Theta_A); \Theta_c); \Theta_D)$$

Blended Latent Diffusion



Self-Attention Guidance

$$Q_t^h = X_t W_Q^h, K_t^h = X_t W_K^h, A_t^h = \text{softmax}(Q_t^h (K_t^h)^T / \sqrt{d})$$

$$M_t = \mathbb{1}(A_t > \psi)$$

$$\hat{x}_t = (1 - M_t) \odot x_t + M_t \odot \bar{x}_t$$

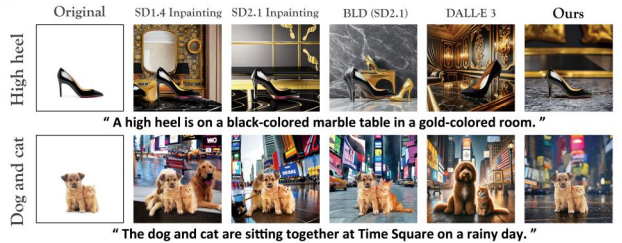
$$\tilde{\epsilon}(x_t) = \epsilon_{\theta}(\hat{x}_t) + (1 + s)(\epsilon_{\theta}(x_t) - \epsilon_{\theta}(\hat{x}_t))$$

Results

I Qualitative result of our method



II Comparison with other models



III Ablation study : ControlNet and Self-Attention Guidance

